

KCET-2016-Set-A-1
Questions – Chemistry

1. The half-life period of a 1st order reaction is 60 minutes. What percentage will be left over after 240 minutes?
 - (1) 6.25%
 - (2) 4.25%
 - (3) 5%
 - (4) 6%

2. Which of the following is not a colligative property?
 - (1) Osmotic pressure
 - (2) Optical activity
 - (3) Depression in Freezing point
 - (4) Elevation in Boiling point

3. The contribution of particle at the edge centre to a particular unit cell is,

(1) $\frac{1}{2}$

(2) $\frac{1}{4}$

(3) 1

(4) $\frac{1}{8}$

4. When an electrolyte is dissociated in solution, the van't Hoff's factor (i) is,

(1) > 1

(2) < 1

(3) $= 0$

(4) $= 1$

5. Which of the following is incorrect in a galvanic cell?

(1) Oxidation occurs at anode.

(2) Reduction occurs at cathode.

(3) The electrode at which electrons are gained is called cathode.

(4) The electrode at which electrons are lost is called cathode.

6. A secondary cell is one
- (1) can be recharged.
 - (2) can be recharged by passing current through it in the same direction.
 - (3) can be recharged by passing current through it in the opposite direction.
 - (4) can not recharged.
7. Osmotic pressure of the solution can be increased by,
- (1) increasing the temperature of the solution.
 - (2) decreasing the temperature of the solution.
 - (3) increasing the volume of the vessel.
 - (4) diluting the solution.
8. The amount of current in Faraday is required for the reduction of 1 mol of $\text{Cr}_2\text{O}_7^{--}$ ions to Cr^{3+} is,
- (1) 1 F
 - (2) 2 F
 - (3) 6 F

(4) 4 F

9. For a chemical reaction,



If the concentration of A is doubled, the reaction rate will be,

- (1) Doubled
- (2) Quadrupled
- (3) Increases by 8 times
- (4) Unchanged

10. Schottky defect in a crystal is observed when,

- (1) Unequal number of cations and anions are missing from the lattice
- (2) Equal number of cations and anions are missing from the lattice
- (3) An ion leaves its normal site and occupies an interstitial site
- (4) No ion is missing from its lattice site

11. $3\text{A} \rightarrow 2\text{B}$, rate of reaction $+\frac{d[\text{B}]}{dt}$ is equal to

$$(1) -\frac{3}{2} \frac{d[A]}{dt}$$

$$(2) -\frac{2}{3} \frac{d[A]}{dt}$$

$$(3) +2 \frac{d[A]}{dt}$$

$$(4) -\frac{1}{3} \frac{d[A]}{dt}$$

12. The activation energy of a chemical reaction can be determined by,

- (1) evaluating rate constants at two different temperatures.
- (2) changing the concentration of reactants.
- (3) evaluating the concentration of reactants at two different temperatures.
- (4) evaluating rate constant at standard temperature.

13. Which of the following statements is incorrect w.r.t. Physisorption?

- (1) The forces involved are van der Waal's forces.

- (2) More easily liquifiable gases are adsorbed easily.
- (3) Under high pressure it results into Multi-molecular layer on adsorbent surface.
- (4) $\Delta H_{\text{adsorption}}$ is low and +Ve.

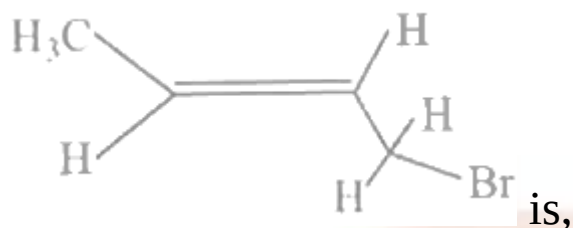
14. Sulphur sol contains

- (1) Discrete S-atoms
- (2) Discrete S-molecules
- (3) Large aggregates of S-molecules
- (4) Water dispersed in Solid Sulphur

15. Reactions in Zeolite catalyst depend on,

- (1) Pores
- (2) Apertures
- (3) Size of cavity
- (4) All of these

16. IUP AC name of the compound



- (1) 1-Bromo but-2-ene
- (2) 2-Bromo-2-butene
- (3) Bromo butene
- (4) 1-Bromo but-3-ene

17. Replacement of Cl of Chlorobenzene to give phenol requires drastic conditions, but Cl of 2, 4-dinitro chlorobenzene is readily replaced. This is because,

- (1) $-\text{NO}_2$ group makes the ring electron rich at ortho and para positions
- (2) $-\text{NO}_2$ group withdraws electrons from meta position.
- (3) $-\text{NO}_2$ donate electrons at meta position.
- (4) $-\text{NO}_2$ withdraws electrons from ortho and para positions.

18. In the reaction,

Ethanol $\xrightarrow{\text{PCl}_5}$ X $\xrightarrow{\text{alc KOH}}$ Y $\xrightarrow[\text{H}_2\text{O}, \Delta]{\text{H}_2\text{SO}_4, \text{Room temp.}}$ Z, The product

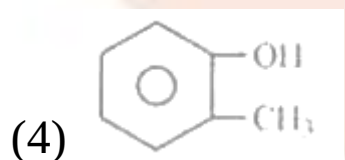
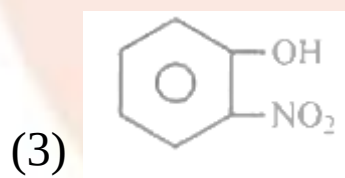
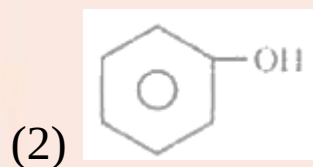
Z is,

- (1) C_2H_4
- (2) $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$
- (3) $\text{CH}_3\text{CH}_2\text{OSO}_3\text{H}$



19. Which of the following compound is most acidic?

- (1) $\text{Cl}-\text{CH}_2-\text{CH}_2-\text{OH}$



20. Benzene carbaldehyde is reacted with concentrated NaOH solution to give the products A and B. The product A can be

used food preservative and the product B is an aromatic hydroxy compound where OH group is linked to sp^3 hybridised carbon atom next to Benzene ring. The product A and B are respectively.

- (1) Sodium benzoate and phenol
- (2) Sodium benzoate and phenyl methanol
- (3) Sodium benzoate and cresol
- (4) Sodium benzoate and picric acid

21. The reaction which involves dichlorocarbene as an electrophile is,

- (1) Reimer-Tiemann reaction
- (2) Kolbe's reaction
- (3) Friedel-Craft's acylation
- (4) Fittig's reaction

22. Ethanol is converted into ethoxy ethane,

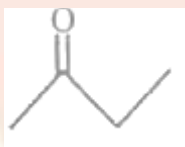
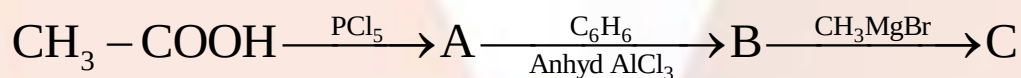
- (1) by heating excess of ethanol with conc. H_2SO_4 at $140^\circ C$.
- (2) by heating Ethanol with excess of cone. H_2SO_4 at 443 K.
- (3) by treating with cone. H_2SO_4 at room temperature.

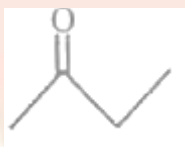
(4) by treating with cone. H_2SO_4 at 273 K.

23. An organic compound X is oxidised by using acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution. The product obtained reacts with phenyl hydrazine but does not answer silver mirror test. The compound X is,

- (1) 2-propanol
- (2) Ethanal
- (3) Ethanol
- (4) $\text{CH}_3\text{CH}_2\text{CH}_3$

24. Predict the product 'C' in the following series of reactions:

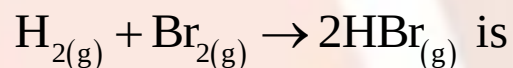


- (1) 
- (2) $\text{CH}_3\text{CH}(\text{OH})\text{C}_6\text{H}_5$
- (3) $\text{CH}_3\text{CH}(\text{OH})\text{C}_2\text{H}_5$
- (4) $(\text{CH}_3)_2\text{C}(\text{OH})\text{C}_6\text{H}_5$

25. The number of oxygen atoms in 4.4 gm of CO_2 is,

- (1) 1.2×10^{23}
- (2) 6×10^{22}
- (3) 6×10^{23}
- (4) 12×10^{23}

26. If the bond energies of H-H, Br-Br and H-Br are 433, 192 and 364 kJ mol^{-1} respectively, then ΔH° for the reaction:



- (1) -261 kJ
- (2) $+103 \text{ kJ}$
- (3) $+261 \text{ kJ}$
- (4) -103 kJ

27. In the reaction; $\text{Fe}(\text{OH})_{3(s)} \rightleftharpoons \text{Fe}_{(aq)}^{3+} + 3\text{OH}_{(aq)}^-$ if the

concentration of OH^- ions is decreased by $\frac{1}{4}$ times, then the

equilibrium concentration of Fe^{3+} will increase by,

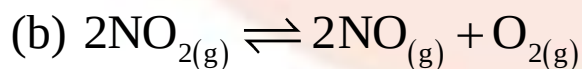
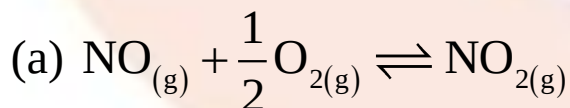
- (1) 8 times

- (2) 16 times
- (3) 64 times
- (4) 4 times

28. The correct statement regarding entropy is

- (1) At absolute zero temperature, entropy of a perfectly crystalline solid is zero.
- (2) At absolute zero temperature, the entropy of a perfectly crystalline substance is +Ve.
- (3) At absolute zero temperature, the entropy of all crystalline substances is zero.
- (4) At 0°C the entropy of a perfect crystalline solid is zero.

29. Equilibrium constants K_1 and K_2 for the following equilibria



(1) $K_1 = \sqrt{K_2}$

(2) $K_2 = \frac{1}{K_1}$

(3) $K_1 = 2K_2$

(4) $K_2 = \frac{1}{K_1^2}$

30. Van-Arkel method of refining Zirconium involves,

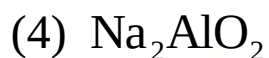
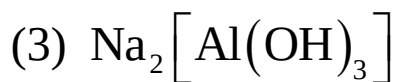
- (1) removing all oxygen and nitrogen impurities
- (2) removing CO impurity
- (3) removing Hydrogen impurity
- (4) removing silica impurity

31. The composition of 'copper matte' is,

- (1) $\text{Cu}_2\text{S} + \text{FeS}$
- (2) $\text{Cu}_2\text{S} + \text{Cu}_2\text{O}$
- (3) $\text{Cu}_2\text{S} + \text{FeO}$
- (4) $\text{Cu}_2\text{O} + \text{FeS}$

32. The complex formed when Al_2O_3 is leached from Bauxite using concentrated NaOH solution is,

- (1) $\text{Na}[\text{Al}(\text{OH})_4]$



33. The property which is **not** true about Fluorine is,

- (1) Most of its reactions are exothermic.
- (2) It forms only one oxo acid.
- (3) Highest electronegativity.
- (4) High F-F bond dissociation enthalpy.

34. Which is true regarding nitrogen?

- (1) Less electronegative
- (2) Has low ionisation enthalpy
- (3) d-orbitals are available
- (4) Ability to form $p\pi-p\pi$ bonds with itself

35. The shape of XeF_6 is,

- (1) Square planar
- (2) Distorted octahedral

(3) Square pyramidal

(4) Pyramidal

36. The number of isomers possible for the octahedral complex

$[\text{CoCl}_2(\text{en})(\text{NH}_3)_2]^+$ is,

(1) Two

(2) Three

(3) No isomer

(4) Four isomers

37. CO is a stronger ligand than Cl^- , because

(1) CO is a neutral molecule.

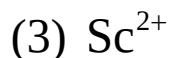
(2) CO has π -bonds.

(3) CO is poisonous.

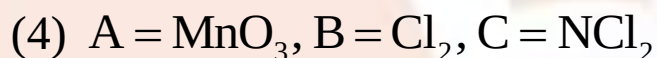
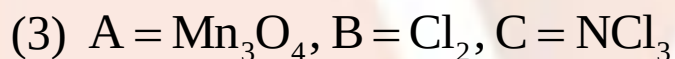
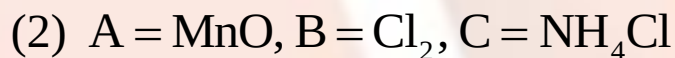
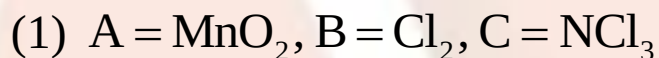
(4) CO is more reactive.

38. The bivalent metal ion having maximum paramagnetic behaviour among the first transition series elements is

(1) Mn^{2+}



39. When a brown compound of Mn (A) is treated with HCl, it gives a gas (B). The gas (B) taken in excess reacts with NH_3 to give an explosive compound (C). The compounds A, B and C are;



40. Mn^{2+} compounds are more stable than Fe^{2+} compounds towards oxidation to their +3 state, because

(1) Mn^{2+} is more stable with high 3rd ionisation energy.

(2) Mn^{2+} is bigger in size.

(3) Mn^{2+} has completely filled d-orbitals.

(4) Mn^{2+} does not exist.

41. Which of the following sequence is correct regarding field strength of ligands as per spectrochemical series?

- (1) $\text{SCN}^- < \text{F}^- < \text{CN}^- < \text{CO}$
- (2) $\text{F}^- < \text{SCN}^- < \text{CN}^- < \text{CO}$
- (3) $\text{CN}^- < \text{F}^- < \text{CO} < \text{SCN}^-$
- (4) $\text{SCN}^- < \text{CO} < \text{F}^- < \text{CN}^-$

42. As per IUPAC norms, the name of the complex $[\text{Co}(\text{en})_2(\text{ONO})\text{Cl}]\text{Cl}$ is

- (1) Chlorido bis(ethane-1,2-diamine) nitro-o-cobalt (III) chloride.
- (2) Chloro bis(ethylene diamine) nitro-o-cobalt (III) chloride.
- (3) Chlorido di(ethylene diamine) nitro cobalt (III) chloride.
- (4) Chloro ethylene diamine nitro-o-cobalt (III) chloride.

43. In the following sequence of reactions ;



The compound A is

- (1) Propane nitrile

(2) Ethane nitrile

(3) CH_3NO_2

(4) CH_3NC

44. An organic compound A on reduction gives compound B, which on reaction with trichloro methane and caustic potash forms C. The compound 'C' on catalytic reduction gives N-methyl benzenamine, the compound 'A' is

(1) Nitrobenzene

(2) Nitromethane

(3) Methanamine

(4) Benzenamine

45. Which of the following gives positive Fehling's solution test?

(1) Sucrose

(2) Glucose

(3) Fats

(4) Protein

46. A liquid can exist only,

- (1) Between triple point and critical point
- (2) At any temperature above melting point
- (3) Between melting point and critical point
- (4) Between boiling and melting points

47. The energy of electron in the n^{th} bohr orbit of H-atom is

- (1) $\frac{-13.6}{n^2} \text{eV}$
- (2) $\frac{-13.6}{n} \text{eV}$
- (3) $\frac{-13.6}{n^4} \text{eV}$
- (4) $\frac{-13.6}{n^3} \text{eV}$

48. Consider the following sets of quantum numbers:

Which of the below setting is not permissible arrangement of electrons in an atom?

n l m s

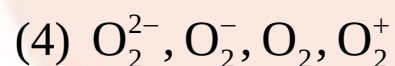
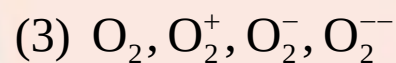
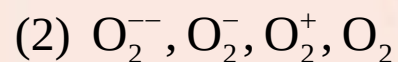
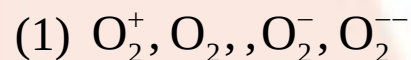
$$(1) \quad 4 \quad 0 \quad 0 \quad -\frac{1}{2}$$

$$(2) \quad 5 \quad 3 \quad 0 \quad +\frac{1}{2}$$

$$(3) \quad 3 \quad 2 \quad -2 \quad -\frac{1}{2}$$

$$(4) \quad 3 \quad 2 \quad -3 \quad +\frac{1}{2}$$

49. The increasing order of bond order of O_2 , O_2^+ , O_2^- and O_2^{2-} is



50. HCl gas is covalent and NaCl is an ionic compound this is because

(1) Sodium is highly electro +Ve.

(2) Hydrogen is a non-metal.

(3) HCl is a gas.

- (4) Electronegativity difference between H and Cl is less than 2.1.

51. Which of the following is not true?

- (1) In vulcanisation the rubber becomes harder and stronger.
- (2) Natural rubber has 'trans' configuration at every double bond.
- (3) Buna-S is a co-polymer of Butene and styrene.
- (4) Natural rubber is 1, 4-polymer of isoprene.

52. Which of the following is a polyamide?

- (1) Nylon-6,6
- (2) Terylene
- (3) Polythene
- (4) Buna-S

53. Which of the following is correct about H-bonding in DNA?

- (1) A-T, G-C
- (2) A-G, T-C
- (3) G-T, A-C

(4) A-A, T-T

54. Which of the following is employed as Tranquilizer?

- (1) Equanil
- (2) Naproxen
- (3) Tetracyclin
- (4) Dettol

55. Reactivity of order of halides for dehydrohalogenation is

- (1) $R-F > R-Cl > R-Br > R-I$
- (2) $R-I > R-Br > R-Cl > R-F$
- (3) $R-I > R-Cl > R-Br > R-F$
- (4) $R-F > R-I > R-Br > R-Cl$

56. Main axis of diatomic molecule is Z. The orbitals P_x and P_y overlap to form

- (1) π -molecular orbital
- (2) σ -molecular orbital
- (3) δ -molecular orbital
- (4) No bond is formed

57. The hybridisation of C in diamond, graphite and ethyne is in the order

(1) sp^3 , sp , sp^2

(2) sp^3 , sp^2 , sp

(3) sp , sp^2 , sp^3

(4) sp^2 , sp^3 , sp

58. A miscible mixture of $C_6H_6 + CHCl_3$ can be separated by

(1) Sublimation

(2) Distillation

(3) Filtration

(4) Crystallisation

59. An organic compound contains C=40%, H=13.33% and N=46.67%. It's empirical formula is

(1) C_2H_2N

(2) C_3H_7N

(3) CH_4N

(4) CHN

60. Electrophile that participates in nitration of benzene is

(1) NO^+

(2) NO_2^+

(3) NO

(4) NO_3^-

KCET-2016-Set-A-1
Answer keys

1	1	16	1	31	1	46	4
2	2	17	4	32	1	47	1
3	2	18	4	33	4	48	4
4	1	19	3	34	4	49	4
5	4	20	2	35	2	50	4
6	3	21	1	36	4	51	2
7	1	22	1	37	2	52	1
8	3	23	1	38	1	53	1
9	2	24	4	39	1	54	1
10	2	25	1	40	1	55	2
11	2	26	4	41	1	56	4
12	1	27	3	42	1	57	2
13	4	28	1	43	2	58	2
14	3	29	4	44	1	59	3
15	4	30	1	45	2	60	2

KCET-2016-Set-A-1

Solutions – Chemistry

1. If the half-life period of a first order reaction is equal to 60 minutes. The total number of half-lives is denoted by n . The total number of half-lives is calculated as follows.

$$T = n \times t_{1/2}$$

$$\begin{aligned} n &= \frac{T}{t_{1/2}} \\ &= \frac{240}{60} \\ &= 4 \end{aligned}$$

So, the total number of half-lives is $n = 4$. The initial amount is supposed to be 100. Then, the remaining amount $(a - x)$

after half-lives $= \frac{100}{2^n}$ is calculated below.

$$\begin{aligned} (a - x) &= \frac{100}{2^4} \\ &= \frac{100}{16} \\ &= 6.25\% \end{aligned}$$

2. The properties which are dependent on the number of solute particles but not on the nature of the solute are known as colligative properties. So, optical activity is not a colligative property because it is independent of number of solute particles.

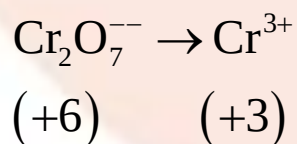
3. The contribution of particle at the edge centre is $\frac{1}{4}$ to a particular unit cell because an atom is shared between 4 unit cells at the edge center of the crystal lattice.

4. According to van't Hoff's factor equation,

$$\text{Van't Hoff's factor} = \frac{\text{Number of particles after dissociation}}{\text{Number of particles taken}}$$

So, if an electrolyte dissociated in the solution then, there is increase in number of particles and the van't Hoff's factor (i) is equal to >1 .

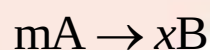
5. The electrode at which electrons are lost is called anode not cathode because at anode oxidation occurs and it acts as a reducing agent.
6. The secondary cell is able to be recharged by passing current through it in the opposite direction because secondary cell can be reused again.
7. As osmotic pressure of the solution is directly proportional to the temperature, So, by increasing the temperature of the solution, osmotic pressure of the solution also increased.
8. The equation for the reduction of $\text{Cr}_2\text{O}_7^{--}$ to Cr^{3+} is given below.



So, the oxidation state of chromium gets changed means there is change in 3 moles of electrons/mole of Cr-atom and $\text{Cr}_2\text{O}_7^{--}$ contains 2 moles of Cr atom. So, there is total

6 moles of electrons/2 mole of Cr-atom. Thus, there is total 6 Faraday is required for the reduction of 1 mol of $\text{Cr}_2\text{O}_7^{--}$ ions to Cr^{3+} .

9. The given reaction is,



So, the rate law for this reaction is given as $r = k[\text{A}]^2$ or $r \propto [\text{A}]^2$

If the concentration gets doubled, then the rate of reaction becomes quadrupled as mentioned below.

$$r \propto [2\text{A}]^2$$

$$r \propto 4[\text{A}]^2$$

10. If equal number of cations and anions are missing from the lattice then it is known as Schottky defect of a crystal. This effect is responsible for decreasing the density of crystal

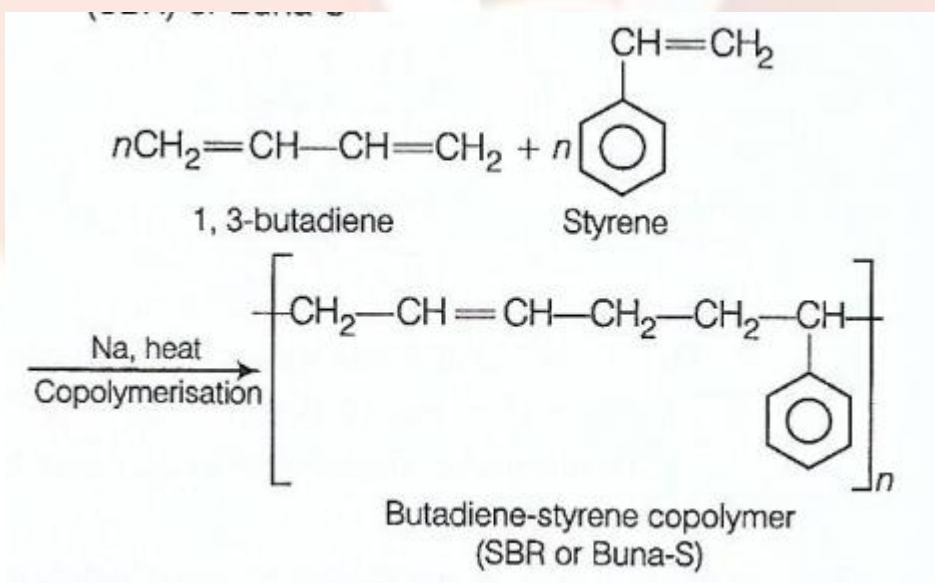
11. The given reaction is, $3A \rightarrow 2B$.

So, the rate of reaction is $-\frac{1}{3} \frac{d[A]}{dt} = +\frac{d[B]}{dt}$

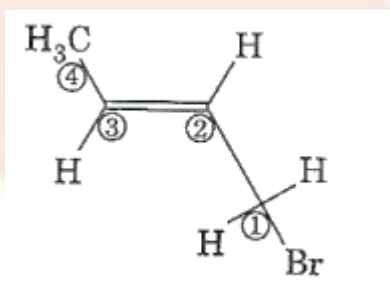
$$\frac{d[B]}{dt} = -\frac{2}{3} \frac{d[A]}{dt}$$

12. The activation energy, E_a , of a chemical reaction is determined by evaluating rate constants at two different temperatures. The equation for the same is shown below.

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left[\frac{T_2 - T_1}{T_2 T_1} \right]$$



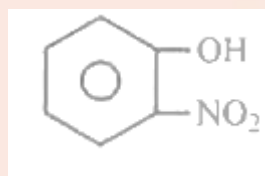
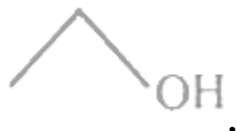
13. With respect to physisorption $\Delta H_{\text{adsorption}}$ is always $-Ve$.
So, the option (4) is incorrect.
14. Sulphur is a type of sol which possesses the large aggregates of sulphur molecules, so they come in the range of colloidal state.
15. The zeolites reaction are dependent upon the pores, aperture and on the cavity size only. So, all of these are correct.
16. The given structure is shown below.

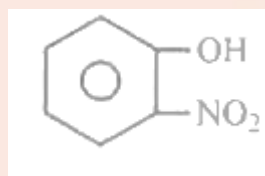


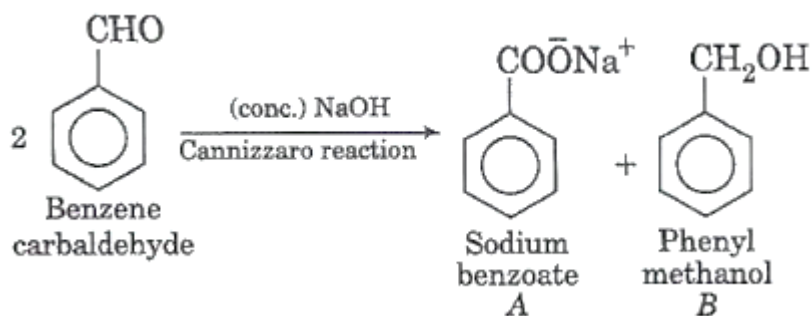
It contains four carbon atoms with a double bond in its skeleton structure and bromo group at 1-position.
Therefore, the IUPAC name of this structure is 1-Bromobut-2-ene.



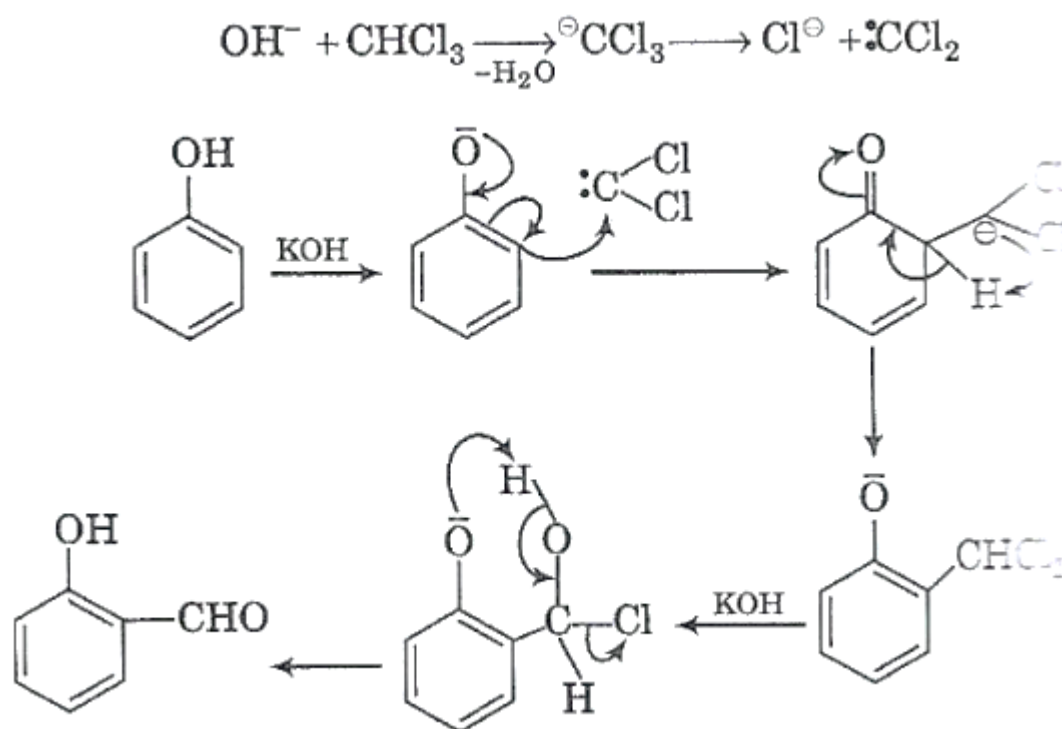
So, the product Z is



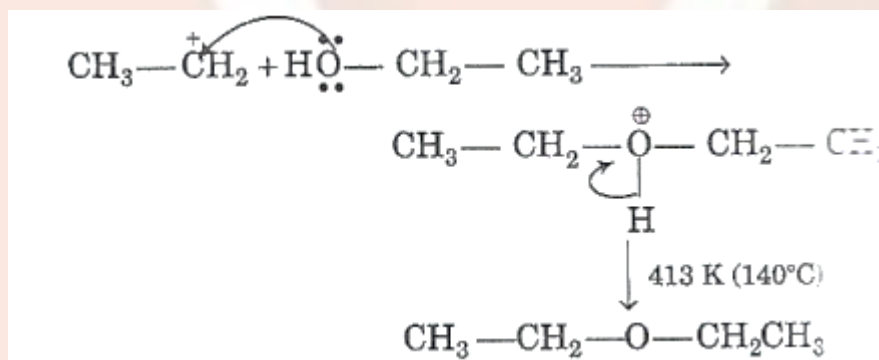
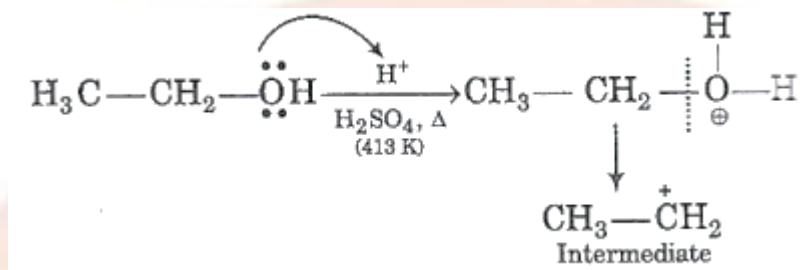
19. The most acidic compound is  because it contains -NO₂ group is electron withdrawing in nature. So, this -NO₂ group weakens the bonding of -OH that result in increasing the acidic nature of the compound.
20. The reaction of benzene carbaldehyde with NaOH moves through Cannizzaro reaction which is a type disproportionation reaction. In this reaction, half aldehyde molecules get oxidized and other molecules get reduced. The structures of compound A and B are shown below.



21. The reaction which involves dichlorocarbene as an electrophile is Reimer-Tiemann reaction in which dichlorocarbene behaves as an electrophile to form salicylaldehyde as shown below.

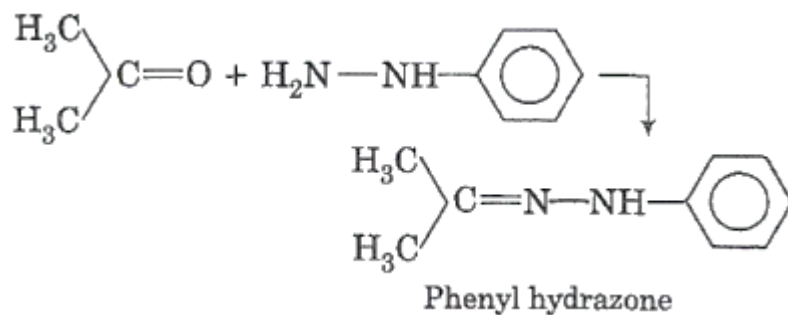


22. The conversion of ethanol into ethoxy ethane is done by heating excess of ethanol with conc. H_2SO_4 at 140°C or 413 K as shown below.

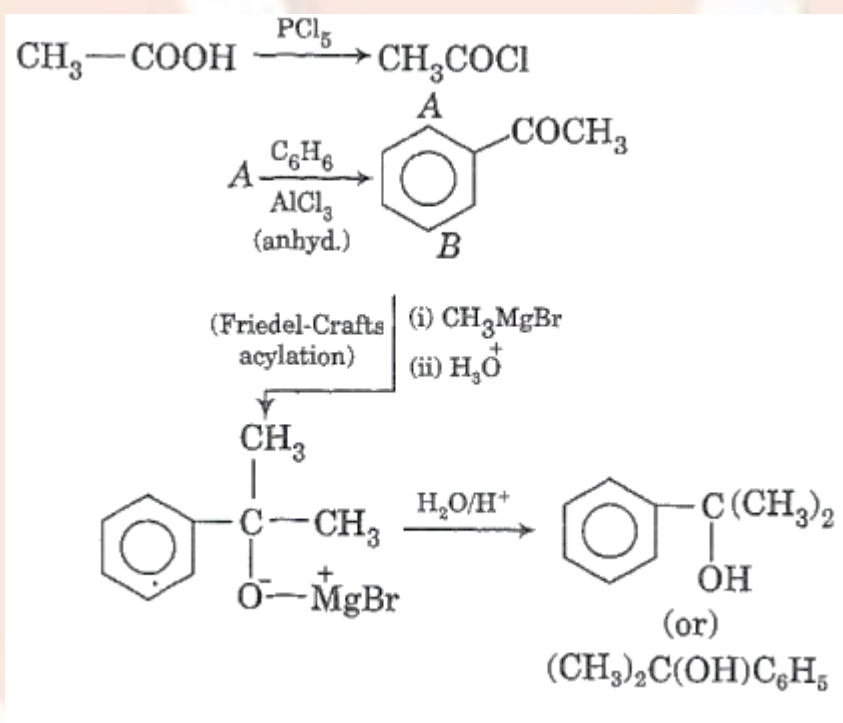


23. If 2-propanol, $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$ gets oxidized by $\text{K}_2\text{Cr}_2\text{O}_7$

solution then it forms acetone, $\text{CH}_3-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{CH}_3$ and it does not show mirror test. Acetone reacts with phenyl hydrazine as shown below.

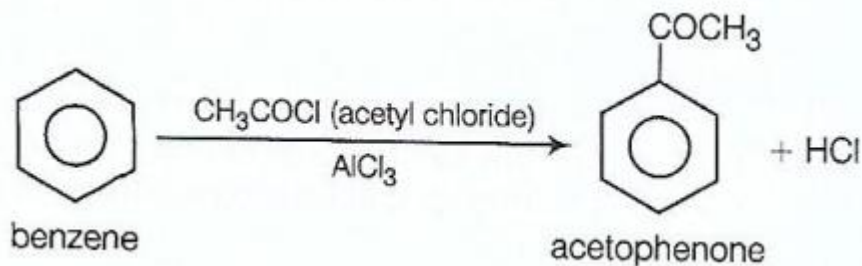


24. The complete reaction is shown below.



So, the compound C is $(\text{CH}_3)_2\text{C}(\text{OH})\text{C}_6\text{H}_5$.

25. The molar mass of CO_2 is 44 g/mol. The number of oxygen atoms in 4.4 gm of CO_2 is calculated as follows.



$$\frac{W}{M} = \frac{N}{N_A}$$

$$\frac{4.4}{44} = \frac{N}{6 \times 10^{23}}$$

$$N = \frac{4.4 \times 6 \times 10^{23}}{44}$$

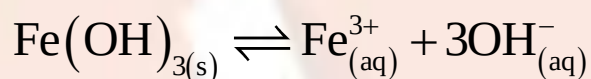
$$= 6 \times 10^{22}$$

If each CO_2 molecule contains 2 atoms then the total number is $2 \times 6 \times 10^{22} = 1.2 \times 10^{23}$.

26. The given reaction is $\text{H}_{2(g)} + \text{Br}_{2(g)} \rightarrow 2\text{HBr}_{(g)}$. The bond energies of H-H, Br-Br and H-Br are 433, 192 and 364 kJ mol^{-1} respectively. The ΔH° of the reaction is calculated below.

$$\begin{aligned}\Delta H^\circ &= \sum \text{BE}(\text{reactants}) - \sum \text{BE}(\text{products}) \\ &= (433 + 192) - (2 \times 364) \\ &= 625 - 728 \\ &= -103 \text{ kJ mol}^{-1}\end{aligned}$$

27. The given reaction is shown below.



The equilibrium constant for the above reaction is expressed as follows.

$$\begin{aligned}K_{\text{eq}} &= \frac{[\text{Fe}^{3+}][\text{OH}^{-}]^3}{[\text{Fe}(\text{OH})_3]} \\ &= \frac{[\text{Fe}^{3+}][\text{OH}^{-}]^3}{1}\end{aligned}$$

If the concentration of $[\text{OH}^{-}]$ decreases by $1/4$ times, then the value of equilibrium constant is,

$$\begin{aligned}K_{\text{eq}} &= [\text{Fe}^{3+}] \left(\frac{1}{4} \cdot [\text{OH}^{-}] \right)^3 \\ &= [\text{Fe}^{3+}] \left(\frac{1}{64} \cdot [\text{OH}^{-}] \right)^3\end{aligned}$$

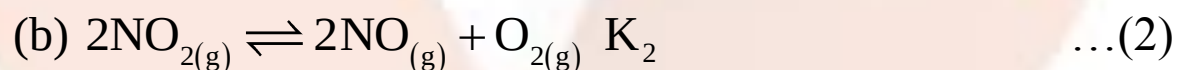
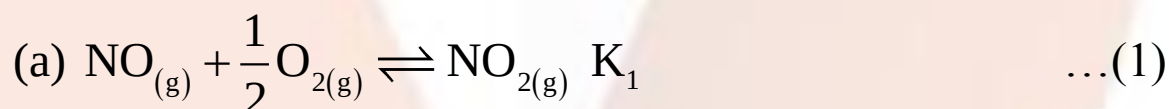
Thus, for the constant value of K_{eq} , the concentration of iron ion should be increased by 64 times as given below.

$$K_{eq} = 64 \left(\frac{1}{64} [\text{Fe}^{3+}] \cdot [\text{OH}^-] \right)$$

$$K_{eq} = [\text{Fe}^{3+}] \cdot [\text{OH}^-]$$

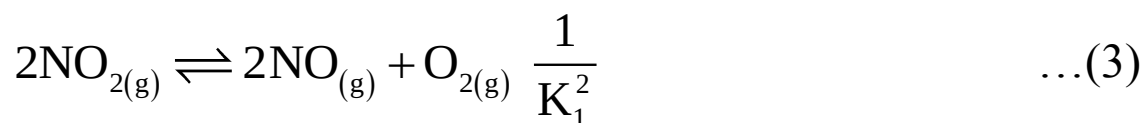
28. According to the third law of thermodynamics, at absolute zero temperature, entropy of a perfectly crystalline solid is zero is a correct statement regarding entropy.

29. The given reactions are,



The comparison of equation (1) and (2) keeps equation (1) equals to equation (2).

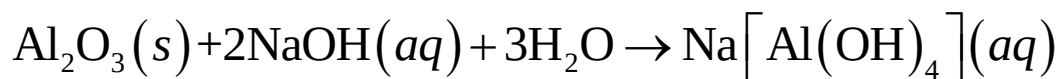
After reversing and multiplying equation (1), the new equation obtained is,



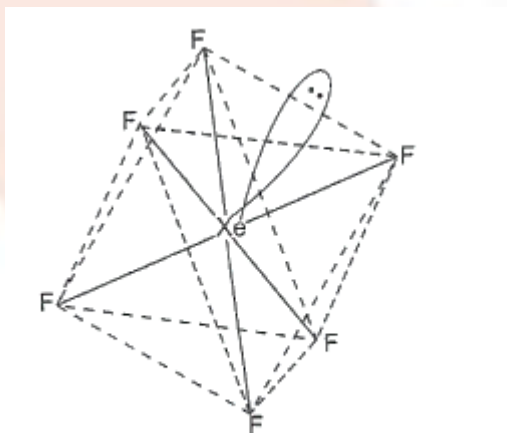
Therefore, the equation (2) is equal to equation (3) as shown below.

$$K_2 = \frac{1}{K_1^2}$$

30. The method of van-Arkel is used for the zirconium refining which includes the elimination of oxygen and nitrogen impurities. This method leads to the heating of the impure metals in the presence of iodine inside a sealed vessel which forms a metallic iodide. Then, the decomposition of metallic iodide into purified takes place at high temperature.
31. The composition, $\text{Cu}_2\text{S} + \text{FeS}$ corresponds to copper matte.
32. The complex formed when Al_2O_3 is leached from Bauxite using concentrated NaOH solution is $\text{Na}[\text{Al}(\text{OH})_4]$ as shown below.

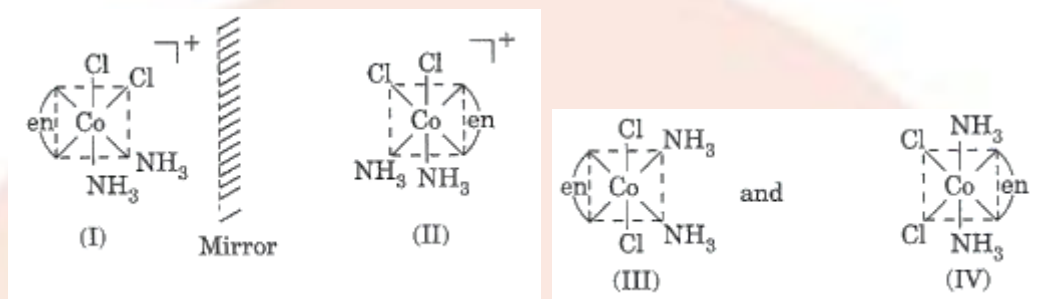


33. The incorrect property about fluorine is that the F-F bond has high dissociation enthalpy as the volume of fluorine atom has 7 valence electrons, thus, fluorine has the low F-F bond dissociation enthalpy.
34. The element nitrogen is able to form the $p\pi-p\pi$ bonds with itself.
35. The shape of XeF_6 is distorted octahedral due to the presence of six sigma bonds and one lone pair of electrons as shown below.

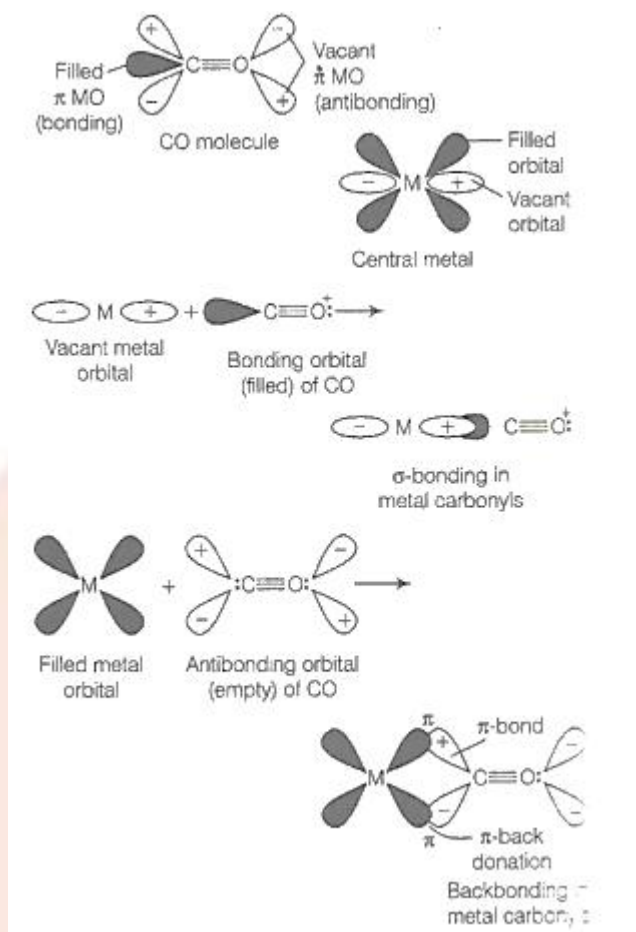


Distorted octahedral

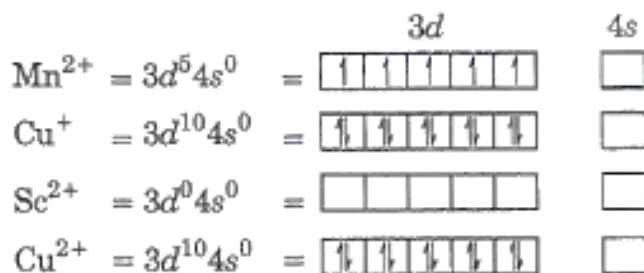
36. The number of isomers possible for the octahedral complex $[\text{CoCl}_2(\text{en})(\text{NH}_3)_2]^+$ is four in which two are optical isomers and two are geometrical isomers as shown below.



37. CO is a stronger ligand than Cl^- , because CO has π -bonds which are present for the donation of pi electrons to another ion or atom as shown below.



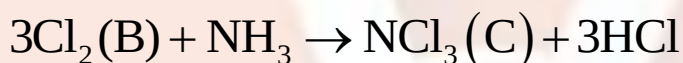
38. The bivalent metal ion which has maximum number of unpaired electrons is Mn^{2+} among the given first transition series elements that corresponds to the maximum paramagnetic behaviour. The orbital diagrams of the first transition series elements are shown below.



39. When a brown compound of Mn (A) is treated with HCl, it gives a gas (B) as shown below.



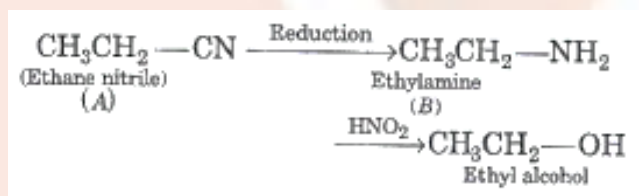
The gas (B) taken in excess reacts with NH_3 to give an explosive compound (C) as shown below.



Therefore, the compounds A, B and C are MnO_2 , Cl_2 , and NCl_3 .

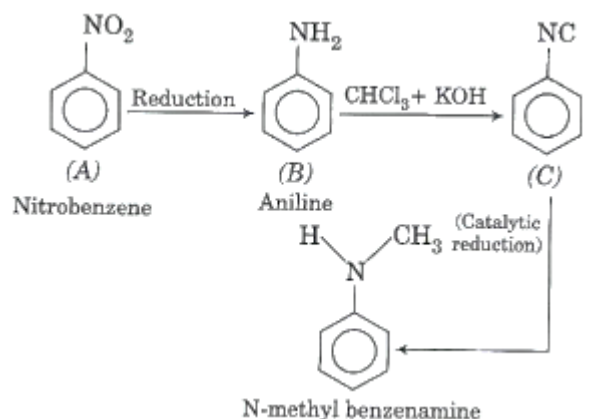
40. The compounds of Mn^{2+} are more stable than Fe^{2+} compounds towards oxidation to their +3 state, because Mn^{2+} is more stable with high 3rd ionisation energy as it has complete half filled stable configuration.

41. The correct sequence regarding field strength of ligands as per spectrochemical series is $\text{SCN}^- < \text{F}^- < \text{CN}^- < \text{CO}$.
42. The IUPAC name of the complex $[\text{Co}(\text{en})_2(\text{ONO})\text{Cl}]\text{Cl}$ is Chlorido bis(ethane-1,2-diamine) nitro-o-cobalt (III) chloride.
43. The complete reaction is shown below.



Therefore, the compound A is ethane nitrile.

44. An organic compound A that is nitrobenzene, which on reduction gives compound B that is aniline, which on reaction with trichloro methane and caustic potash forms C. The compound 'C' on catalytic reduction gives N-methyl benzenamine as shown below.



Therefore, the compound 'A' is nitrobenzene.

45. Glucose gives positive Fehling's solution test because aldehyde group is present in glucose.

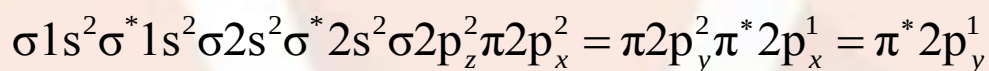
46. A liquid can exist only between boiling and melting points as these points brief about the temperature over which the change of phase takes place in a species.

47. As, $E_n = \frac{-13.6 Z^2}{n^2} \text{eV}$ and for hydrogen $Z=1$, so the energy

of electron in the n^{th} bohr orbit of H-atom is $\frac{-13.6}{n^2} \text{eV}$.

48. The quantum number's set, $n = 3$, $l = 2$, $m = -3$ and $s = +\frac{1}{2}$ because if $l = 2$ then $m \neq -3$. So, this set of setting is not permissible arrangement of electrons in an atom.

49. The electronic configuration of oxygen molecule with atomic number 16 is given below.



The bond order of O_2 is given below.

$$\begin{aligned} \text{Bond order} &= \frac{\text{Bonding electrons} - \text{antibonding electrons}}{2} \\ &= \frac{10 - 6}{2} \\ &= 2 \end{aligned}$$

The bond order of O_2^+ with atomic number 15 is given below.

$$\begin{aligned} \text{Bond order} &= \frac{\text{Bonding electrons} - \text{antibonding electrons}}{2} \\ &= \frac{10 - 5}{2} \\ &= 2.5 \end{aligned}$$

The bond order of O_2^- with atomic number 17 is given below.

$$\begin{aligned}\text{Bond order} &= \frac{\text{Bonding electrons} - \text{antibonding electrons}}{2} \\ &= \frac{10 - 7}{2} \\ &= 1.5\end{aligned}$$

The bond order of O_2^{2-} with atomic number 18 is given below.

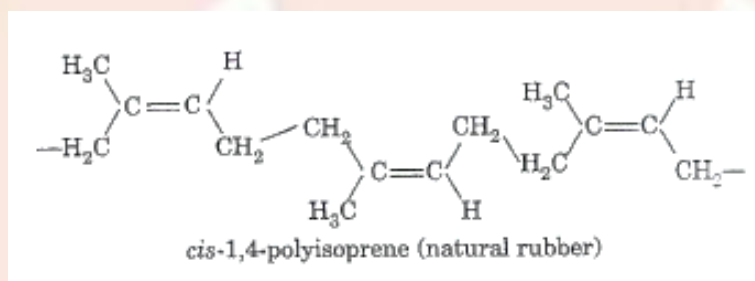
$$\begin{aligned}\text{Bond order} &= \frac{\text{Bonding electrons} - \text{antibonding electrons}}{2} \\ &= \frac{10 - 8}{2} \\ &= 1\end{aligned}$$

Thus, the increasing order of bond order of O_2, O_2^+, O_2^- and O_2^{2-} is $O_2^{2-}, O_2^-, O_2, O_2^+$.

50. The electronegativity of Cl is 3 and Na is 0.88. The difference in their electronegativity is 2.12 but the electronegativity of Cl is 3 and H is 2.2. The difference in their electronegativity is less than 2.12. So, HCl gas is covalent and NaCl is an ionic compound this is because the

electronegativity difference between H and Cl is less than 2.1.

51. Natural rubber has 'cis' configuration at every double bond not 'trans' configuration. Natural rubber cis-1,4 polymer of isoprene as shown below.



So, option (2) is an incorrect statement.

52. Nylon-6,6 is a polyamide of adipic acid, $\text{HOOC}-(\text{CH}_2)_4-\text{COOH}$ and hexamethylene diamine $\text{NH}_2-(\text{CH}_2)_6-\text{NH}_2$. The structure of nylon-6,6 is shown below.



53. The hydrogen bonding in DNA takes place between Adenine-Thymine(A-T), and Guanine-Cytosine (G-C).
54. The compounds used for treating the metal diseases are known as tranquilizers. Equanil is a tranquilizer used to treat depression.
55. Reactivity of order of halides for dehydrohalogenation is $R-I > R-Br > R-Cl > R-F$ because the greater bond length corresponds to the easy dehydrohalogenation.
56. Main axis of diatomic molecule is Z. The orbitals P_x and P_y overlap to form no bond because main-axis is z, so x and y axes gets perpendicular to z-axis.
57. Diamond possesses tetrahedral structure with 4 sigma bonds, graphite possesses hexagonal structure with 3 sigma bonds and ethyne possesses only 1 sigma bond. So, the

hybridisation of C in diamond, graphite and ethyne is in the order sp^3 , sp^2 , sp respectively.

58. A miscible mixture of $C_6H_6 + CHCl_3$ can be separated by distillation as C_6H_6 and $CHCl_3$ has high difference in their boiling points.
59. An organic compound contains C=40%, H=13.33% and N=46.67%. It's empirical formula is calculated as follows in the table.

Element	% by mass	Atomic mass	Relative no. of moles of elements	Simplest molar ratio
C	40	12	$\frac{40}{12} = 3.33$	$\frac{3.33}{3.33} = 1$
H	13.33	1	$\frac{13.33}{1} = 13.33$	$\frac{13.33}{3.33} = 4$
N	46.67	14	$\frac{46.67}{14} = 3.33$	$\frac{3.33}{3.33} = 1$

So, the empirical formula is CH_4N .

60. Electrophile that participates in nitration of benzene is NO_2^+ which is formed by concentrated sulfuric acid and nitric acid as shown below.

